

Algebraic paradoxes in adaptive quantum computation

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We show that if an adaptive $\mathbb{Z}_2$ -linear measurement-based quantum computing protocol deterministically computes a non-affine Boolean function, then the underlying quantum resource satisfies an inconsistent set of linear equations. This witnesses an algebraic form of strong contextuality generalising Mermin’s All-versus-Nothing arguments. Such algebraic contextuality can be detected cohomologically, resolving an open question posed by Raussendorf, who had established cohomological witnesses of contextuality for non-adaptive protocols, but left the adaptive case open. We prove this result constructively: we model adaptive measurement protocols as ordinary measurements on a larger scenario of tree-like measurements, and explicitly build the inconsistent equations inductively.

Introduction A central goal in quantum computing is to delineate the scope of advantage afforded by the use of quantum resources. Rather than focusing on particular algorithms, this line of work seeks general principles that separate quantum and classical computation. Among the candidates, contextuality [7, 13] – the impossibility of reconciling quantum measurement statistics with pre-existing outcome values – has emerged as a leading explanation for quantum advantage across multiple computational models [3, 8, 11, 12, 16]. It is also being explored for benchmarking quantum hardware, enabling error detection beyond standard randomised benchmarking [9]. At the heart of contextuality lies a logical tension – a “paradox”: each empirically-accessible context of compatible measurements provides a consistent classical snapshot, yet combining these into a global description leads to a contradiction [4].

Measurement-based quantum computation (MBQC) [19] provides a particularly clean framework for investigating this question. Here, a classical controller performs successive local measurements on a resource state and processes outcomes, treating the quantum system as a black box that responds to measurement queries. Crucially, later measurement settings may depend on prior outcomes, a feature known as *adaptivity*.

Such adaptivity is essential for computational universality, but it complicates the analysis of nonclassicality. Existing mathematical tools for contextuality, especially algebraic [2] and cohomological [2, 5, 17, 18] witnesses, have largely been limited to non-adaptive “flat” scenarios. This work addresses a fundamental open question: can the contextuality required for adaptive quantum computation always be witnessed by simple, algebraic contradictions? The answer provided is yes, and the result significantly strengthens our understanding of contextuality as a computational resource.

Overview of result Raussendorf [16] considered $\mathbb{Z}_2$ -linear MBQC, where the classical controller is restricted to linear operations modulo 2. He showed that if such a (possibly adaptive) protocol computes a non-affine Boolean function deterministically, then it must exploit strong contextuality in its resource state. We strengthen this by showing the contextuality can always be witnessed via an All-versus-Nothing (AvN) argument. These are elegant algebraic proofs that establish logical contradictions through systems of linear equations satisfied by quantum measurement outcomes but violated by any classical value assignment; classic examples include the GHZ paradox [10] and the Peres–Mermin magic square [14, 15]. This in turn implies that the contextuality is cohomologically nontrivial, resolving an open question posed by Raussendorf for the adaptive case [17, 18].

The main technical challenge is treating adaptivity. Rather than developing a new theory of contextuality for adaptive processes, the paper introduces a method for reducing adaptive measurement protocols to non-adaptive ones. The central idea, drawing on [1], is a systematic “flattening” construction that models entire adaptive measurement protocols as single measurement objects in an enlarged measurement scenario. Adaptive branching is encoded into a tree-like structure, whose outcomes are algebraically well defined. In this way, adaptivity is not ignored, but absorbed into the structure of the measurement scenario itself. This transformation preserves the operational behaviour of the original computation, while allowing standard algebraic tools to be applied. In particular, it makes it possible to define a linear theory of measurement outcomes and to reason about global consistency in the usual way. Importantly, the construction is explicit. Given an adaptive computation, the corresponding algebraic contradiction can be systematically derived inductively from the structure of the protocol, rather than inferred indirectly.

An AvN argument: the GHZ paradox To illustrate AvN arguments and motivate our results, we recall the GHZ paradox and the related Anders–Browne construction [6]

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The GHZ paradox involves measurements at three sites, with two settings available at each: x_0, x_1 at the first, y_0, y_1 at the second, and z_0, z_1 at the third, each yielding outcomes in $\mathbb{Z}_2$ (viewed as a ring; the algebraic structure is essential for the argument). A shared GHZ state $|\psi_{\text{GHZ}}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ with X - and Y -basis measurements at each site produces the behaviour (“empirical model”). For each of four measurement contexts, the observed joint outcomes always satisfy a linear equation over $\mathbb{Z}_2$:

$$\begin{aligned} x_0 + y_0 + z_0 &= 0 & x_0 + y_1 + z_1 &= 1 \\ x_1 + y_0 + z_1 &= 1 & x_1 + y_1 + z_0 &= 1 \end{aligned} \tag{1}$$

This system is inconsistent: summing all four equations yields $0=1$. Therefore, no classical value assignment to all six measurements is consistent with the observed outcomes, establishing strong contextuality. Algebraic contradictions of this form are known as *All-versus-Nothing* (AvN) arguments [2].

Computing with AvN paradoxes: the Anders–Browne construction Anders and Browne [6] showed how this enables a parity computer – restricted to affine operations over $\mathbb{Z}_2$ – to compute non-affine Boolean functions, thus enhancing its computational power.

Consider the non-affine OR function $(a, b) \mapsto a \vee b = a + b + ab$. Given input $(a, b) \in \mathbb{Z}_2^2$, linear preprocessing encodes it as $(a, b, a+b)$, determining measurement context $\{x_a, y_b, z_{a+b}\}$. The three outcomes are then summed (linear postprocessing), yielding a single output bit:

$$x_a + y_b + z_{a+b} = a \vee b \quad (a, b \in \mathbb{Z}_2), \tag{2}$$

deterministically computing OR. Observe that eq. (2) precisely matches the equations in (1): the GHZ AvN argument underlies this computation.

Since OR together with affine operations is functionally complete, one can compute arbitrary Boolean functions by composing such constructions. This requires *adaptivity*: later measurements depend (affinely) on prior outcomes. For instance, composing two Anders–Browne constructions to compute a three-wise OR produces:

$$x'_{x_a+y_b+z_{a+b}} + y'_c + z'_{x_a+y_b+z_{a+b}+c}$$

where primed variables denote measurements on a second resource state. The output is no longer a simple linear expression in the measurement outcomes. With greater adaptive depth, such expressions become increasingly complex. This raises our central question: for arbitrary adaptive $\mathbb{Z}_2$ -linear MBQC protocols computing non-affine functions, can we always find AvN arguments witnessing the exploited contextuality? We answer this affirmatively by extending AvN arguments to adaptive protocols, which we treat as first-class measurements.

Setup: Linear measurement protocols We formalise adaptive $\mathbb{Z}_2$ -MBQC via linear measurement protocols operating on a Bell-type scenario B over multiple sites, each with two measurement settings (indexed by $\mathbb{Z}_2$) yielding outcomes in $\mathbb{Z}_2$. A protocol proceeds through (some of) the sites in a fixed order, performing one measurement per site (thus determining a context), with later choices depending linearly on past outcomes.

We represent such a protocol as a binary tree, where a node specifies a measurement with a branch for each possible outcome. Formally, a protocol is either empty or has the form (x, v, q) ,

$$\begin{array}{c} x \\ 0/ \setminus 1 \\ q \quad v.q \end{array}$$

where x is the initial measurement, q is the continuation protocol (upon obtaining outcome 0), and $v \in \mathbb{Z}_2^{s(q)}$ specifies the update to measurement settings at the sites of q upon obtaining outcome 1. The adaptivity structure may be encoded by a strictly lower-triangular matrix T_p whose (i, j) -entry indicates how the outcome at site i affects the measurement setting at site j .

Computing with linear protocols We study the use of such protocols with access to a resource state to compute a Boolean function $f: \mathbb{Z}_2^l \rightarrow \mathbb{Z}_2^o$. An MBQC on scenario B consists of: an empirical model e on B (modelling the resource), a base linear protocol p using sites $s(p)$ (corresponding to input $\mathbf{0}$), and linear pre- and post-processing maps $Q: \mathbb{Z}_2^l \rightarrow \mathbb{Z}_2^{s(p)}$ and $Z: \mathbb{Z}_2^{s(p)} \rightarrow \mathbb{Z}_2^o$. Given input $w \in \mathbb{Z}_2^l$, we encode it via Q to obtain a vector that updates the default measurement settings of p , yielding the linear protocol $Q(w).p$. Running this on e produces a vector of measurement outcomes in $\mathbb{Z}_2^{s(p)}$ (one per site), which we postprocess via Z to obtain the output. Such an MBQC (e, p, Q, Z) computes f deterministically if every input $w \in \mathbb{Z}_2^l$ always yields $f(w)$, regardless of the (possibly probabilistic) outcomes obtained from e .

Main result To apply existing tools to the adaptive setting, following [1], we construct a measurement scenario $\text{MP}(B, \mathbb{Z}_2)$ whose measurements are themselves measurement protocols. We integrate post-processing into the protocol to ensure all measurements have outcomes in the ring $\mathbb{Z}_2$, a requirement for AvN arguments, despite protocols visiting different numbers of sites. Moreover, to establish the AvN argument, we must work in a scenario containing a broader class of protocols beyond the linear ones. Most importantly, we include multiplicative nodes, which halt and return 0 unless a specific outcome is observed. These enable us to isolate branches and express protocols as sums of smaller subprotocols, which may in turn satisfy additional equations. In addition, it will be convenient to consider higher-arity branches, which measure the parity of a set of measurements (at different sites) in a single step. Any model e on B extends naturally to $\text{MP}(e, \mathbb{Z}_2)$.

Standard reductions allow focusing on single-output Boolean functions with two inputs: a non-affine function $\mathbb{Z}_2^l \rightarrow \mathbb{Z}_2^o$ reduces to a non-affine coordinate function, which in turn restricts to a non-affine function on some two-dimensional subspace.

Theorem. *If an MBQC (e, p, Q, Z) deterministically computes a non-affine function $f: \mathbb{Z}_2^2 \rightarrow \mathbb{Z}_2$, then the model $\text{MP}(e, \mathbb{Z}_2)$ has an inconsistent $\mathbb{Z}_2$ -linear theory, an All-versus-Nothing argument witnessing strong contextuality.*

Proof sketch An MBQC (e, p, Q, Z) deterministically computes $f: \mathbb{Z}_2^2 \rightarrow \mathbb{Z}_2$ iff

$$\text{MP}(e, \mathbb{Z}_2) \models Q(w).p; Z = f(w) \quad \text{for all } w \in \mathbb{Z}_2^2, \quad (3)$$

where $Q(w).p; Z$ denotes the measurement in $\text{MP}(B, \mathbb{Z}_2)$ that runs $Q(w).p$ and post-processes by Z . When f is non-affine, and thus unbalanced, summing these four equations yields 1 on the right-hand side. Hence, establishing an AvN argument reduces to showing that

$$\sum_{w \in \mathbb{Z}_2^2} Q(w).p; Z = 0 \quad (4)$$

can be derived from the $\mathbb{Z}_2$ -linear theory of $\text{MP}(e, \mathbb{Z}_2)$. In fact, this equation holds for any empirical model e , so we can safely suppress e from the notation.

We prove eq. (4) by induction on $n := |s(p)|$, the number of sites in p . The base case $n=1$ is immediate: $Q: \mathbb{Z}_2^2 \rightarrow \mathbb{Z}_2$ cannot be injective, so each term appears an even number of times and thus cancels. For $n > 1$, we employ three key reductions:

1. If $\text{Rank} T_p = n$, we show that there exists another protocol q such that

$$\sum_{w \in \mathbb{Z}_2^2} Q(w).p; Z = \sum_{w \in \mathbb{Z}_2^2} Q(w).q; Z$$

and $\text{Rank} T_q < n$. Thus we may assume T_p is not of maximal rank.

2. If $\text{Rank} T_p < n$, we show that we can condense p to have strictly fewer levels without affecting the value of $\sum_{w \in \mathbb{Z}_2^2} Q(w).p; Z$. The cost is that we need to allow higher-arity nodes, which measure the parity $\sum x_i$ of a set of measurements in one step.
3. We show that we can systematically replace all higher-arity branching nodes by sums of protocols with unary nodes. This rewrites the overall sum as a sum of smaller sums $\sum_w Q_i(w).p_i; Z_i$, where each p_i has strictly fewer than n sites, reducing to the induction hypothesis. The cost is that we need to allow measurement protocols with *multiplicative nodes* which, instead of branching, continue only if a specific outcome is observed, otherwise returning 0. While such nodes enable non-affine functions, it turns out that each increases the input dimension by 1, and the result holds provided the input dimension is sufficiently large relative to the number of multiplicative nodes.

Significance This work strengthens the conceptual foundations of quantum computation by making precise the connection between adaptivity, contextuality, and computational power. It clarifies the role of contextuality in adaptive MBQC and brings adaptive protocols within the scope of well-understood non-adaptive contextuality frameworks.

By systematically “flattening” adaptive protocols into an equivalent non-adaptive scenario, we show that the contextuality forced by deterministic computation of non-affine Boolean functions in $\mathbb{Z}_2$ -linear MBQC can always be witnessed in AvN form. Hence, it is also cohomologically detectable, which resolves Raussendorf’s open question about whether cohomological witnesses extend to adaptive computation.

From a computational perspective, the result provides a sharp necessity statement: non-affine classical computation in this model occurs if and only if global classical value assignments are obstructed by explicit algebraic inconsistency. In this sense, quantum computational power is directly linked to the failure of classical logical explanations. The construction is explicit and inductive: given an adaptive computation, the corresponding algebraic contradiction can be systematically derived from the protocol’s structure.

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